SEAN BALBALE

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EDUCATION

Trinity College — Hartford, CT *Expected May 2027*

B.S. Electrical & Computer Engineering | B.S. Computer Science | GPA: 3.65 / 4.00

Coursework: High-Performance Computing, Deep Learning, Software Engineering, Embedded Systems Design, Microelectronic Circuits, Data Structures & Algorithms, Intro to Computer Systems, Applied Linear Algebra, Calculus III.

Extracurriculars: Varsity Athlete — Men's Rowing (Div. III); Technology Chair & Singer — The Trinity College Accidentals.

Northeastern University — Boston, MA & London, UK Attended Sept. 2023 – Dec. 2024; all credits transferred to Trinity

TECHNICAL SKILLS

Languages: Java, Python, C, C++, TypeScript, Assembly (8051), Verilog, SQL, MATLAB, LaTeX
Backend/Cloud: Spring Boot 3, Jakarta EE, JUnit 5, Mockito, REST APIs, AWS, Kubernetes, Docker, CI/CD, Log4j2
Frontend: React, Next.js, Tailwind CSS, Node.js, AngularJS
Embedded/HW: Arduino, 8051 MCU, nRF52840 (Cortex-M4F), Intel Cyclone V FPGA, FreeRTOS, BLE 5, ANT+, LTspice
Tools: Git, Linux, SolidWorks, 3D Printing (Klipper), Slurm/MPI, Precision Metrology, Oscilloscopes

WORK EXPERIENCE

Sonos — Boston, MA *Summer 2026*

Software Engineering Intern — Cloud Accounts (SW Services)

- Architecting and maintaining backend Java microservices powering user-identity, subscription entitlement, player-settings, and API-permission systems across the Sonos platform.
- Working in an AWS and Kubernetes environment; applying containerization, orchestration, and GitHub source-control best practices on cloud services used globally.

Bullhorn — Boston, MA *Summer 2025*

Software Engineering Intern

- Owned framework migration (BH Boot 1.0.0): upgraded multiple production Java applications from Spring Boot 2 to Spring Boot 3, refactoring all `javax` namespaces to `jakarta` and restructuring the proprietary data access layer.
- Led a critical security initiative upgrading core logging infrastructure to Log4j2; migrated legacy test suites from TestNG/JMockit to JUnit 5 and Mockito across affected services.
- Participated in the full SDLC — managed feature branches, authored merge requests, and conducted peer code reviews (SEARCH_AI, ATS_CRM, MIDDLE_OFFICE teams).

East Coast Metrology — A Division of In-Place Machining Co. — Topsfield, MA *Summer 2024*

Metrology Intern

- Calibrated portable CMMs, laser trackers, total stations, and large-volume scanners; verified manufactured parts against SolidWorks CAD models for dimensional compliance.

Titan Advanced Energy Solutions — Salem, MA *Summer 2023*

Engineering Intern

- Built Python tooling to parse ultrasound JSON data and render waveform visualizations for non-destructive EV battery defect detection, accelerating QA workflows.

Cherrybrook Networks (Founder) — Weston, MA *Mar. 2019 – May 2025*

Freelance Software Developer & IT Consultant

- Operated a six-year freelance business delivering software dev, custom PC builds, Wi-Fi infrastructure, and automated cloud-backup solutions to local clients.

SELECTED PROJECTS

Edge-Compute Pacing & Feedback Node *(Senior Capstone — CPSC 498 / ECE 483)* *Fall 2026 – Spring 2027*

- Embedded edge node providing real-time fatigue feedback for rowers/cyclists via the W′ balance model in C/C++ on FreeRTOS on a Nordic nRF52840 (Cortex-M4F, 64 MHz); targets ≤100 ms sensor-to-LED latency entirely offline via BLE 5 / ANT+.

Cherrybrook Networks — Full-Stack Portfolio *2024 – Present*

Production Next.js / React / TypeScript web app (Vercel CI/CD); custom backend API for transactional email; 151+ commits, 95.5% TypeScript.

Haptic Horizon — Wearable Navigation Device *2025*

Embedded C++ firmware (Arduino): HC-SR04 ultrasonic sensor + haptic motor encodes proximity as vibration patterns for visually impaired users; interrupt-driven real-time event loop.

HONORS & RESEARCH

Alumni Prize in English Composition — Trinity College *May 2026*

"Embodied Harms and Inferred Data: Redefining Privacy in Extended Reality" — accepted for publication in *The International Journal of Law, Ethics, and Technology* (Expected Late May 2026). Proposes a "Bodyright" framework for biometric data autonomy in immersive XR environments; awarded by the Trinity Department of English at Honors Day.

PERSONAL

Competitive varsity rower (Heavyweight eight/four, 2K erg sprints); avid cook — samosas and katsu ramen; weightlifting, running, ski mountaineering. Manage live sound and recording for a collegiate a cappella ensemble.